

PES UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #38 : Dropshipping Inventory & Order
Orchestrator

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SECTION : I

REQUIREMENTS TABLE :

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse inbound Shopify/WooCommerce order webhooks, verify supplier API inventory, and auto-dispatch supplier purchase orders.	High	Pass: Supplier PO is triggered with customer delivery address after successful inventory verification. Fail: Order is placed for an out-of-stock supplier SKU.	Rationale
FR-002	Functional	The system shall split a multi-vendor customer order into separate supplier purchase orders based on the supplier	High	Pass: An order containing products from two suppliers generates two separate supplier purchase orders. Fail: Products from different suppliers are combined into	Rationale

		assigned to each SKU.		one supplier purchase order.	
FR-003	Functional	The system shall record and update shipment tracking numbers received from supplier partners.	High	Pass: A valid tracking number is stored against the corresponding supplier order. Fail: The tracking number is missing or associated with the wrong order	Rationale
FR-004	Functional	The system shall notify the E-Store Owner when a supplier SKU is unavailable or when a supplier purchase order cannot be dispatched.	High	Pass: The E-Store Owner receives a notification when inventory verification or purchase-order dispatch fails. Fail: No notification is generated for a failed supplier order.	Rationale
FR-005	Functional	The system shall maintain the status of each supplier purchase order from creation through	High	Pass: The supplier purchase order status is updated correctly after each valid order-status event. Fail: The status remains incorrect	Rationale

		shipment tracking.		after a valid status event.	
NFR - 001	Nonfunctional	Inbound webhook processing must acknowledge receipt with HTTP 200 within 200 ms to prevent provider webhook retry storm.	High	Pass: Webhook acknowledgement occurs within 200 ms under simulated peak load. Fail: Webhook acknowledgement exceeds 200 ms under the specified test conditions.	Rationale
NFR - 002	Nonfunctional	The system shall authenticate and validate incoming storefront webhooks before processing order information.	High	Pass: Requests with valid authentication are accepted and processed. Fail: Requests with invalid or missing authentication are rejected and no order is processed.	Rationale